MAE 440 Section 01 Aerodynamics Laboratory

Experiment No.2

Pressure Drag of a Smooth Sphere

Submitted By:

Toni Ensley

Maryam Fareed

Michelle Lee

Ravleen Maan

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Abstract:

The experiment is dedicated to the study of pressure drag over a smooth sphere often used to study the patterns seen in a fluid flow. Using a wind tunnel, a sphere is used as the object of interest due to its smooth surface allowing for a more consistent collection of data. A digital manometer records the differential pressure (ΔP) using P_∞ as a reference point. With this setup, data is obtained at each increment of 5° and 10° degrees recording ΔP. Utilizing the data, the pressure coefficient(Cp) is calculated for each reference value providing graphical representations of the flow characteristics around the sphere.

Through the data collected some theories on flow separation, and the Reynolds number's effect on the pressure coefficient can be inferred. At high speeds it is seen that flow separation of the boundary layer is lower than at low speeds. Meaning that the flow is connecting more smoothly in the back minimizing drag when at higher speeds. At higher speeds it is expected for flow to be deemed turbulent, however, turbulent flow does not always mean increased flow separation and drag. Applying this understanding of boundary layer, Cp, and Re correlations allows for points in which manipulation of these values could offer better aerodynamic performance for when mitigating flow separation or drag is desirable.

Background/Theory:

Previously covered, drag force is an aerodynamic force that slows objects as they move through fluids. It is a total of the pressure drag and viscous drag. Pressure drag is the force that pushes an object further back as a result of a pressure differential. The fluid's viscosity causes the flow to separate at the beginning of an object which causes a build-up of stagnation pressure. With the stagnation pressure at the fore front, there is a negative sum of pressure toward the aft end of the object causing a pressure differential. Reynolds number is an indication of the flow's pattern, $Re = \frac{\rho U \infty D}{\mu}$. With a large Reynolds number, the flow is expected to behave turbulently because of the high pressure differential. With more bluff bodies, there will be a larger wake and this increases the proportion of pressure drag. As a result, the pressure coefficient is highly dependent on the Reynolds number because it (Re) resembles how the pressure is distributed on the object. The pressure coefficient has a major influence on the drag coefficient and this will display how the object interacts in various flow states.

Experimental Procedure:

The purpose of the lab is to observe the pressure drag on a smooth sphere. There are two major components for this lab procedure: a digital manometer to automatically record pressure on the computer and a rotary knob. The only sample tested was a smooth, plastic sphere. The wind tunnel was set-up ahead of time with everything in place e.g. sample, pitot tube, computer program. The rotary knob was reset to 0° and the wind tunnel was prompted to start. One student will record the pressure differentials using the computer and another student will turn the rotary knob in 10° intervals. Before each turn, the computer will record approximately 20 seconds of data and then the student will prompt their partner to turn the knob 10°. After the knob is turned 10°, the student will click record and the cycle repeats. The process will repeat until the sphere has completed a full rotation of 360°.

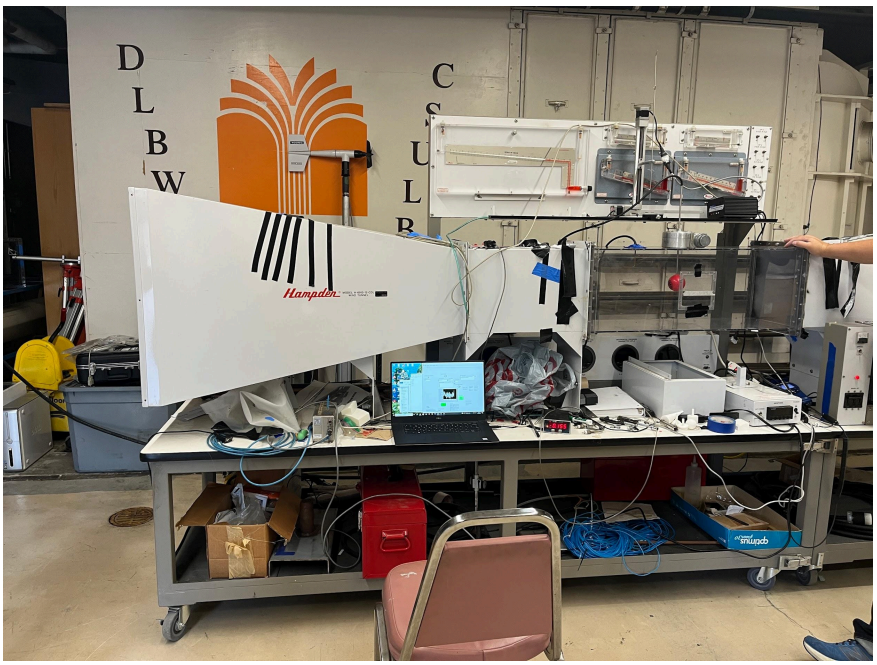


Fig. 1: Wind Tunnel Set-up



Fig. 2: Close-up View of Smooth Sphere

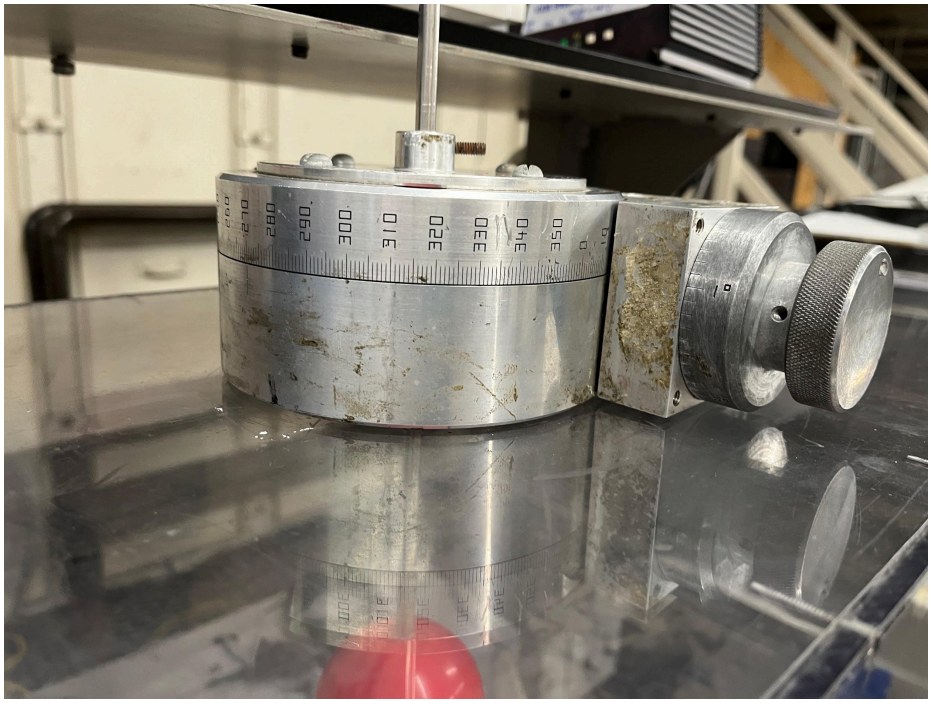


Fig. 3: Close-up View of Rotary Knob

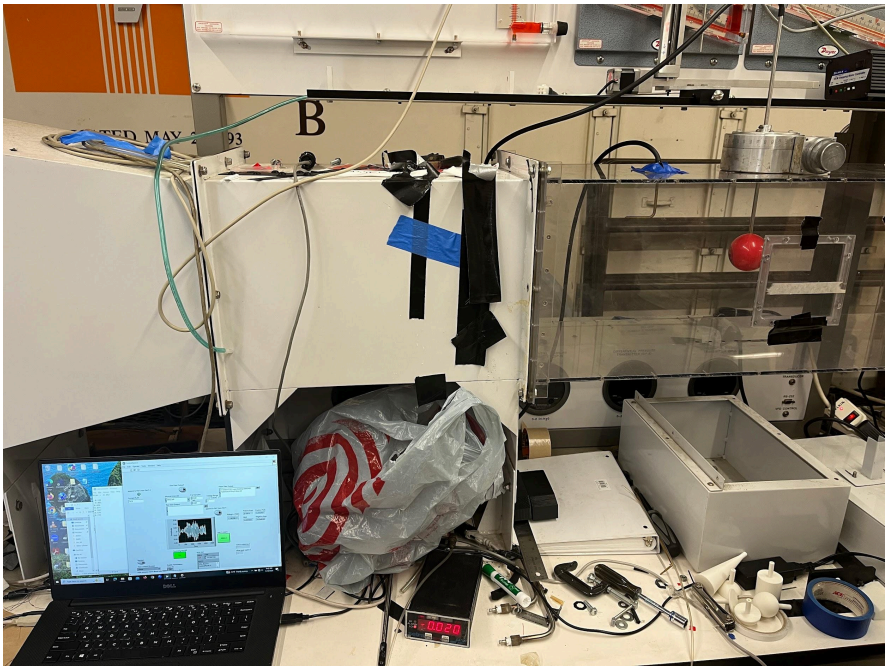


Fig. 4: Close-up View of Experiment Set-up

Raw Data:

D=		2.73"		Voltage	V
				Pressure	Inches of water column

High

Degree	Raw Data Voltage
0	-0.14489
10	-0.13534
20	-0.11458
30	-0.06717
40	0.01217
50	0.10524
60	0.18589
70	0.24771
80	0.28741
90	0.25082
100	0.21129
110	0.21452
120	0.16752
130	0.14707
140	0.14342
150	0.13058
160	0.13265
170	0.12103
180	0.10612
190	0.12215
200	0.12336
210	0.13132
220	0.13169
230	0.14084
240	0.1448
250	0.19093
260	0.21007
270	0.25518
280	0.28607
290	0.25589
300	0.20354
310	0.12255
320	0.03501
330	-0.03841
340	-0.10364
350	-0.13865
360	-0.14953

Low

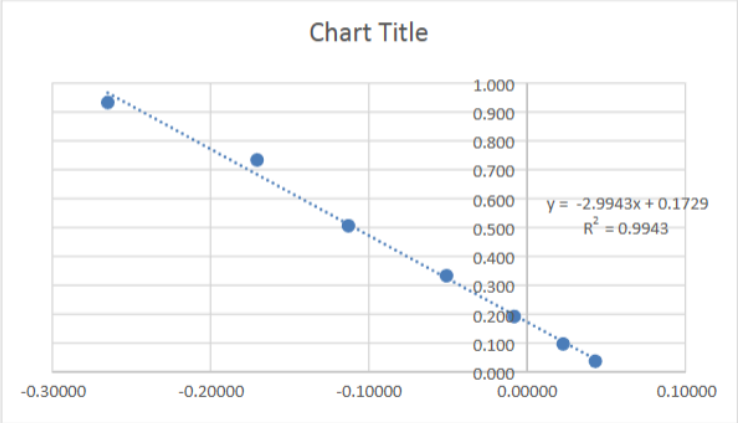
Degree	Raw Data Voltage
0	-0.00587
10	0.00065
20	0.01092
30	0.02941
40	0.05321
50	0.07415
60	0.09371
70	0.10407
80	0.09552
90	0.09172
100	0.08804
110	0.08456
120	0.08587
130	0.08673
140	0.08662
150	0.08388
160	0.08391
170	0.07843
180	0.07692
190	0.08176
200	0.084
210	0.08631
220	0.07946
230	0.08014
240	0.08917
250	0.08445
260	0.08399
270	0.1006
280	0.09651
290	0.09465
300	0.08002
310	0.05681
320	0.03624
330	0.01458
340	0.00116
350	-0.00336
360	-0.0055

Low(Refined)

Degree	Raw Data Voltage
0	-0.00134
5	-0.0027
10	-0.00165
15	0.0019
20	0.01165
25	0.02208
30	0.03101
35	0.04065
40	0.05241
45	0.06464
50	0.07321
55	0.08302
60	0.0952
65	0.10253
70	0.10584
75	0.10488
80	0.09314
85	0.08826
90	0.09157
95	0.08444
100	0.08474
105	0.08442
110	0.08707
115	0.08132
120	0.08417
125	0.08764
130	0.08659
135	0.08828
140	0.08806
145	0.08512
150	0.08543
155	0.08355
160	0.08316
165	0.08426
170	0.0778
175	0.07527
180	0.07685
185	0.08013
190	0.0823
195	0.0839
200	0.08039
205	0.0844
210	0.08397
215	0.08444
220	0.08176
225	0.0803
230	0.08197
235	0.08638
240	0.08004
245	0.08939
250	0.0825
255	0.08749
260	0.08412
265	0.08261
270	0.08628
275	0.10461
280	0.10737
285	0.09961
290	0.09477
295	0.08672
300	0.07969
305	0.06874
310	0.05726
315	0.04608
320	0.03436
325	0.02334
330	0.01499
335	0.00815
340	0.00199
345	-0.00134
350	-0.00265
355	-0.00238
360	-0.00153

	Voltage	Bias fix	Bias
-0.121	0.04314	0.037	
-0.061	0.02278	0.097	
0.034	-0.00812	0.192	
0.175	-0.05090	0.333	
0.348	-0.11289	0.506	
0.576	-0.17061	0.734	
0.775	-0.2650	0.933	

-0.158 equation in form Pressure = -2.9943*voltage+0.1729



Sample Calculations and Results:

Term Bank
$C_p = \ Pressure\ coefficient$
$P = \ Absolute\ Pressure$
$P_\theta = \ Circumferential\ Pressure$
$P_\infty = \ Static\ Pressure$
$P_0 = \ Stagnation\ Pressure$
$F_d = \ Drag\ force$
$A = \ Frontal\ projected\ \ area$
$U_\infty = \ freestream\ velocity$
$\mu = \ dynamic\ viscosity$
D = diameter of sphere

Calculation of Cp (pressure coefficients) at each speed:

Pressure equation from raw data: $P_\theta = (-\ 2.9943\ *\ Voltage) + 0.1729$

Convert Pressure Units

Inches of water column ⇒ Pascals

1 in of water = 249.09

Pressure Coefficient Equation

$$C_p = \frac{P_\theta - P_\infty}{P_0 - P_\infty} = \frac{\Delta P_\theta}{\Delta P_0}$$

Same logic is applied to the remaining Cp values for all speeds

$P_\theta = (-\ 2.9943\ *\ (-\ 0.14489) + 0.1729$

$P_\theta = \ 0.606744127\ \ Units: inchs\ of\ water\ column$

$P_{\theta} = 0.606744127 * 249.09 = 151.133 \; Units: Pa$

$C_p = \frac{151.133}{151.133} = 1$

$C_p = \frac{144.01}{151.133} = 0.9528$

where $\Delta P_0 = 151.133$

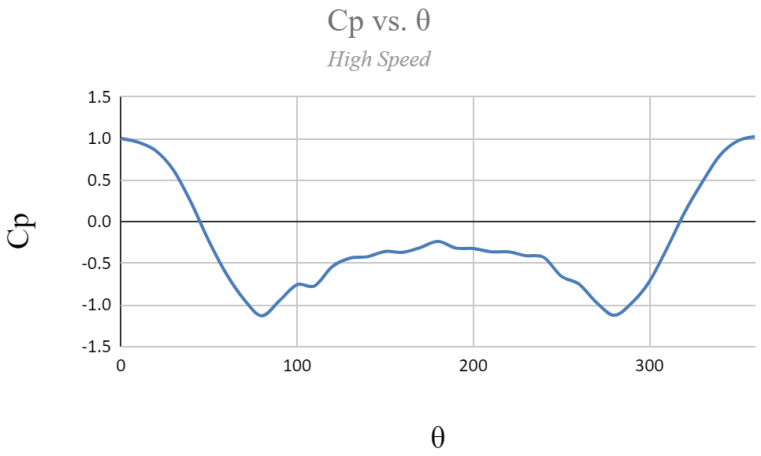


Figure 1A - High Speed Cp vs. θ

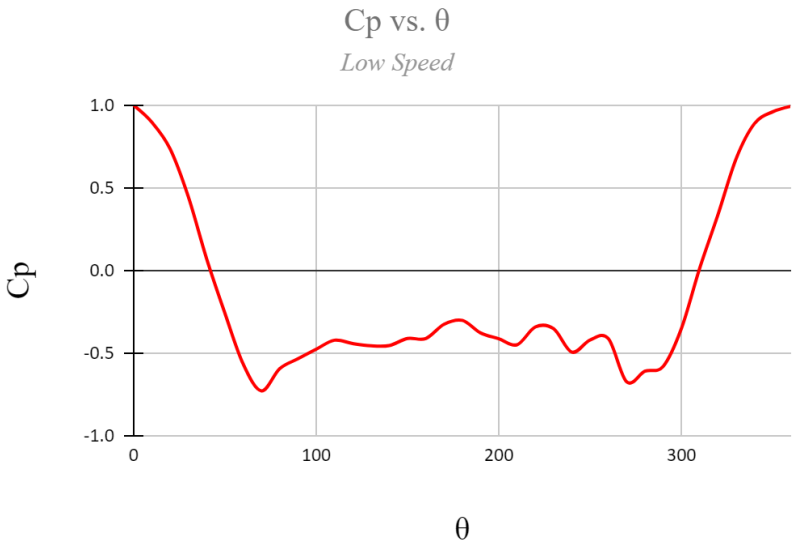


Figure 2A - Low Speed Cp vs. θ

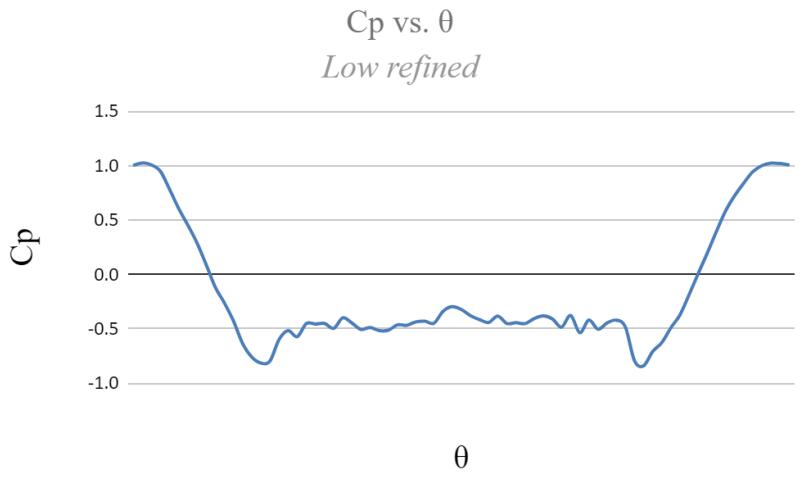


Figure 3A - Low Speed (Refined) Cp vs. θ

Find Freestream Velocity

$$U_\infty = \sqrt{\frac{2\Delta P_0}{\rho}}$$

Same logic is applied to the remaining P_0 values for all speeds

$$U_\infty = \sqrt{\frac{2(151.133\ Pa)}{1.225\ kg/m^2}} = 15.70 m/s^2$$

High	$U_\infty = 15.70\ m/s$
Low	$U_\infty = 8.80\ m/s$
Low (Refined)	$U_\infty = 8.48\ m/s$

Find Re#

$Re_D = \frac{\rho U_\infty D}{\mu}$ where μ is $1.86 * 10^{-5}$Pa*s

$Re_D = \frac{(1.225\ kg/m^3)(15.70\ m/s)(0.069\ m)}{(1.86*10^{-5} Pa*s)}$

High	$Re_D = 71346.3$
Low	$Re_D = 39990.3$
Low(Refined)	$Re_D = 38536.12$

Same logic is applied to the remaining speeds

Drag Coefficient Equation:

$$C_d = \frac{F_d}{1/2\ \rho U_\infty^2 A} = \int_0^{2\pi} \frac{1}{2} C_p(\theta) cos(\theta) d\theta = \sum_{i=1}^{n} \frac{1}{2} C_{pi} cos(\theta i) \Delta\theta$$

Using all Cp Values find Cd

$\Delta\theta = 0\ 1745329252$ ⇒**Incremental change in radians (10 deg)**

Same logic is applied to the remaining Cp values for all speeds

$C_{d2} = \frac{1}{2}(0.9528704709) cos(10^\circ) * (0.1745329252)$

Cd2 = - 0.1395436959

Repeat for all Cp values to find each incremental Cd and sum all these values to get … Cd = 0.03899434521 (drag coefficient High Speed)

Cd(high)			Cd (low)
0.1745329252	Radians	Radians	0.1745329252
-0.1395436959	0.1745329252	0.1745329252	-0.1314357097
0.06057005414			0.05242498819
0.01659602755			0.01199095271
-0.02617945782			-0.00829481543
-0.03947706785			-0.04343813695
0.1051237133			0.0939853489
-0.1036257804			-0.08049846393
0.02183657385			0.01144134337
0.07451554969			0.04177005121
-0.1140449858			-0.07168000933
0.1349036989			0.07350471189
-0.07698377862			-0.06283112649
0.02825922865			0.02921086745
0.01459779326			0.01567251398
-0.04386840592			-0.05014401979
0.06294686122			0.07004372074
-0.05113069605			-0.05323878774
0.02493680464			0.03148806744
-0.003678394584			-0.004369256848
-0.02753464373			-0.03509714847
0.05601456554			0.06927670073
-0.06344302261			-0.05935080116
0.05637814909			0.0484037228
-0.02442849593			-0.02809051125
-0.0276455194			-0.017658407
0.09580368419			0.05258337536
-0.1674014374			-0.1157487032
0.1812727478			0.09803950449
-0.09598915052			-0.05695174628
0.002774861958			0.001350556758
0.02932482956			-0.001344870947
0.01769503596			0.05331604026
-0.0820901249			-0.117382684
0.1055963757			0.1179346733
-0.04797891104			-0.04755007676
-0.05064722123			-0.04922544526
0.03899434521			0.006319672152

Figure 1B - High Speed and Low Speed Cd

Figure 2B - Low Speed (Refined) Cd

Cd (Low Refined)		
0.0872664626	Radians	
0.02532399866	0.0872664626	
-0.07360699344		
-0.06265977199		
0.02778224947		
0.05221141336		
0.006090629583		
-0.02281521032		
-0.005253457972		
-0.005351400637		
-0.0220445554		
-0.0008260893181		
0.05269170782		
0.0372068524		
-0.04499090603		0.02818655115
-0.06417409737		0.03443829324
0.005771242107		-0.01072892516
0.04436975161		-0.04669727573
0.02238704528		-0.008812065043
-0.02879203306		0.0378867384
-0.03438485308		0.02844772305
0.009494322869		-0.01645623038
0.04327389716		-0.04149117863
0.0113458324		-0.007643424559
-0.03177990336		0.06756322407
-0.0347840626		0.0391646752
0.01564930857		-0.03075885377
0.04492704608		-0.04076305654
0.008857806083		0.000716282936
-0.03574001901		0.0156734791
-0.02859516917		-0.0003748167442
0.01856923827		0.01148860594
0.03662628408		0.03121133233
0.002598146056		-0.007836583715
-0.02778752763		-0.06259106361
-0.01549329083		-0.02989386497
0.01688928086		0.06255717149
0.03101592743		0.07322423527
-0.002405013203		-0.02530047153
-0.03769238168		-0.08880255673
-0.01629635996		-0.02483633099
0.02753053364		-0.0005773323616
0.03423926521		
-0.007798973418		
-0.03533449739		
-0.01223795354		

Same logic is applied at all speeds:

Found values so far…

High	**Cd = 0.038**
Low	**Cd = 0.006**
Low (Refined)	**Cd = 0. 0005**

Calculating Fd at each speed:

Assuming…

$\rho = 1.225\ kg/m^3$

Convert 2.73 in to 0.069

$A = \frac{1}{4}\pi d^2$

$A = 0.003\ m^2$

$F_d = C_d \frac{1}{2} \rho A U_\infty$

where U_∞ is at constant low and high speeds

$$F_d = 0.038\frac{1}{2}(1.225kg/m^3)(0.003m^2)(15.70m/s)^2$$

High	Fd = 0.017 N
Low	Fd = $8.53*10^{-4}$ N
Low (Refined)	Fd = $6.60*10^{-5}$ N

Pressure(Pa)	Cp (high)
151.1338946	1
144.0110253	0.9528704709
128.5271754	0.8504192641
93.16641806	0.6164495285
33.99066422	0.2249043096
-35.42561268	-0.2343988605
-95.57843026	-0.6324089677
-141.6868888	-0.9374924745
-171.2971412	-1.133413135
-144.0064829	-0.9528404154
-114.523025	-0.75775871
-116.9321211	-0.7736988544
-81.87716233	-0.5417524808
-66.624526	-0.4408311331
-63.90217282	-0.4228182764
-54.32545642	-0.3594525011
-55.86936631	-0.3696680116
-47.20258713	-0.312322972
-36.08196084	-0.2387416862
-48.03793934	-0.3178502048
-48.94041807	-0.3238215901
-54.87738556	-0.3631044228
-55.15335013	-0.3649303836
-61.97787934	-0.410085901
-64.93144608	-0.4296286167
-99.3375152	-0.657281515
-113.6130878	-0.7517379744
-147.2583897	-0.9743571428
-170.297702	-1.126800196
-147.7879434	-0.9778610136
-108.7426861	-0.7195122335
-48.33627942	-0.3198242164
16.95544595	0.1121882421
71.71576668	0.4745180879
120.3675744	0.7964300444
146.4797894	0.9692054176
154.5946395	1.022898536

Pressure (Pa)	Cp (low)
47.4458016	1
42.58285838	0.8975052996
34.92297696	0.7360604265
21.132207	0.4453967746
3.38097255	0.07125967811
-12.23713037	-0.2579180866
-26.82596002	-0.5654021878
-34.55296796	-0.7282618651
-28.17594886	-0.5938554712
-25.34171815	-0.5341192961
-22.59698946	-0.4762695265
-20.00143081	-0.4215637662
-20.97849456	-0.4421570266
-21.61992572	-0.4556762662
-21.5378822	-0.4539470611
-19.49425269	-0.4108741349
-19.51662819	-0.4113457363
-15.42936917	-0.3251998838
-14.30313538	-0.3014626142
-17.91305029	-0.3775476372
-19.58375471	-0.4127605404
-21.30666864	-0.4490738468
-16.19759486	-0.3413915313
-16.70477299	-0.3520811626
-23.43980017	-0.4940331786
-19.91938729	-0.4198345611
-19.57629621	-0.41260334
-31.96486781	-0.6737133052
-28.91434055	-0.6094183168
-27.5270592	-0.5801790311
-16.61527096	-0.3501947571
0.6959118765	0.01466751226
16.03805022	0.3380288599
32.19316527	0.6785250578
42.20247478	0.8894880761
45.57371763	0.9605426844
47.16983703	0.994183583

Figure 1C - High Speed and Low Speed Cp and P(Pa)

	Pressure (Pa)	Cp (Low Refined)
2	44.06710025	1
1	45.0814565	1.023018448
5	44.29831381	1.005246852
3	41.65054564	0.9451619328
5	34.37850632	0.7801399712
5	26.59928887	0.6036087857
7	19.9388467	0.4524655942
5	12.7488509	0.2893054189
7	3.977652699	0.09026354529
2	-5.144095088	-0.1167332331
3	-11.53603119	-0.261783306
5	-18.85282152	-0.4278207873
5	-27.9372768	-0.6339712993
9	-33.40435867	-0.7580339637
2	-35.87312279	-0.814056804
4	-35.15710661	-0.7978084878
2	-26.40082542	-0.5991051208
3	-22.7610765	-0.5165095133
1	-25.22984062	-0.5725323536
2	-19.91192879	-0.4518547551
2	-20.13568385	-0.4569323539
5	-19.89701179	-0.4515162485
1	-21.87351478	-0.4963683714
5	-17.58487621	-0.3990477274
1	-19.71054924	-0.4472849161
2	-22.29864939	-0.5060158091
7	-21.51550669	-0.4882442133
4	-22.77599351	-0.5168480199
3	-22.61190647	-0.5131244475
5	-20.41910692	-0.4633639791
9	-20.65032048	-0.4686108312
5	-19.24812212	-0.4367912119
3	-18.95724055	-0.4301903334
3	-19.77767576	-0.4488081958
4	-14.95948355	-0.3394705679
1	-13.07248258	-0.2966494846
5	-14.25092587	-0.323391505
9	-16.69731448	-0.3789065854
9	-18.31580939	-0.4156345502
7	-19.50916969	-0.4427150772
7	-16.89123553	-0.383307171
2	-19.88209478	-0.4511777419
1	-19.5613792	-0.4438998503
2	-19.91192879	-0.4518547551
3	-17.91305029	-0.4064948723
9	-16.82410902	-0.3817838914

Pressure (Pa)	Cp (Low Refined)
-18.06967883	-0.4100491915
-21.35887815	-0.4846898941
-16.63018797	-0.3773833058
-23.60388722	-0.5356351355
-18.46497943	-0.4190196161
-22.18677186	-0.5034770097
-19.67325673	-0.4464386497
-18.54702295	-0.4208814023
-21.28429313	-0.4829973611
-34.95572706	-0.7932386489
-37.01427358	-0.839952558
-31.22647613	-0.7086120019
-27.61656122	-0.6266934077
-21.61246722	-0.4904445061
-16.3691404	-0.3714594405
-8.202080854	-0.1861270836
0.3602792924	0.00817569775
8.698884383	0.1974008803
17.44024857	0.3957657408
25.65951764	0.5822828707
31.8873667	0.7236093711
36.98898198	0.8393786241
41.58341913	0.9436386531
44.06710025	1
45.044164	1.022172182
44.84278445	1.017602343
44.20881179	1.003215813

Figure 2C - Low Speed (Refined) Cp and P(Pa)

Discussion of Results:

The purpose of this experiment is to examine how pressure changes around a sphere at different flow speeds. Our sphere had a diameter of 2.73 inches, and we took pressure measurements as the sphere rotated 360 degrees in 10-degree intervals. The high-speed and low-speed data were recorded, and a refined low-speed data table was created using 5-degree intervals. The raw data table shows that the voltage range for the high-speed case is from -0.14953 to 0.28741 V. The range for the low-speed case is -0.00587 to 0.10060 V, and -0.00279 to 0.10584 V for the refined case.

The pressure equation used was Ptheta=-2.9943V+0.1729. Our pressure and Cp results illustrated in Figures 1A, 2A, and 3A show the flow behavior around the sphere for high-speed, low-speed, and refined data, respectively. The pressure coefficient was around 1 at the front stagnation point and became negative around the sides and back side. For the high-speed case, the pressure ranged from -171.30 to 154.59 Pa, while the Cp value ranged from -1.13 to 1.02. For the low-speed case, the pressure ranged from -34.55 to 47.45 Pa, while the Cp values were approximately -0.73 to 1. Lastly, as shown in Figure 3A, the refined low-speed data ranged from -37.01 to 45.05 Pa, with Cp values ranging from -0.81 to 1.02. The negative pressure at the back of the sphere indicates flow separation.

The next section of the data calculates the freestream velocities, which were 15.70 m/s for the high-speed case, 8.8 m/s for the low-speed case, and 8.48 m/s for the refined low-speed case. The Reynolds numbers for these cases are 71,346.3 for the high-speed case, 39,990.3 for the low-speed case, and 38,536.12 for the refined low-speed case. These results tell us that the high-speed condition produces a larger pressure difference because of the dynamic pressure increase. The higher Reynolds number also affects the flow separation around the sphere.

The drag coefficient calculations in Figures 1A and 1B for the high-speed case reported a Cd value of 0.038 and a drag force of 0.017 N, as presented in the summary table. The recorded Cd values for low speed and refined speed are 0.006 and -0.0005, respectively. The drag force values for low speed and refined speed were 8.528*10^-4 N and 6.60*10^-5 N, respectively. A negative drag value is not physically possible. The negative result just tells us that the drag value is approximately zero, and the negative value was likely a mistake in rounding or numerical calculation.

Overall, the high-speed case had the largest velocity, Reynolds number, pressure difference, drag coefficient, and drag force. These results prove that the increase in velocity and Reynolds number results in an increase in the pressure drag on the sphere.

Conclusions and Recommendations:

In this experiment, the pressure distribution and resulting pressure drag on a smooth sphere were investigated across varying freestream velocities (8.48 m/s to 15.50 m/s) and Reynold number (38,536 to 71,346). Pressure coefficients consistently reached a peak of approximately 1.0 at the forward stagnation point (theta = 0, 360) and dropped to negative values along the sides and rear of the sphere, confirming flow separation and the formation on a low-pressure wake. As the freestream velocity increased, the dynamic pressure rose, producing a wider range of differential pressures (-171.30 Pa to 154.59 Pa at high speed) and yielding the highest total pressure drag force ($F_d = 0.017N$, $C_d = 0.038$). Conversely, the lower speeds yielded significantly smaller drag forces (8.53 x $8.53 \times 10^{-4}N$ for low speed and $6.60 \times 10^{-5}N$ for low refined). The minor negative total Cd calculated in the refined low-speed run (-0.0005) is physically non-realizable and represents numerical integration/rounding sensitivity when total pressure drag approaches zero. Overall, the experiment data validates the fundamental aerodynamic relationship between velocity, Reynolds number, surface pressure distribution, and pressure drag on a bluff body.

To improve accuracy of the experiment, it is recommended to re-zero and calibrate the digital manometer prior to each trial to eliminate initial voltage biases. Another thing that can be done is to ensure precise alignment of the sphere's pressure tap relative to 0 degrees to eliminate flow asymmetry between the upper and lower surfaces.

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Group Contributions:

Michelle Lee - Background/Theory, Experimental Procedure

Toni Ensley - Abstract, Raw Data, Sample Calculations and Results

Maryam Fareed - Discussion of Results

Ravleen Maan - Conclusions and Recommendations